(Please write your Enrollment Number) Enrollment No. 11001172022

MID-TERM EXAMINATION

(Course Name: B. Tech.) (Semester: 5)

(October 2024) OFF LINE mode

Subject Code: BCS 303	Subject: Theory of Computation
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
a)	How many tuples are used to define a Grammar? Explain with a suitable example.	2.5	CO 1
b)	What is DFA? How is it different from NFA?	2.5	CO 1
c)	Define Pushdown Automata (PDA) using tuples and explain with an example.	2.5	CO 1
d)	What is Greibach Normal Form (GNF) and how is it different from Context Free Grammar (CFG)? Discuss with a suitable example.	2.5	CO 1
	UNIT I		
Q2	**Attempt any two parts**	(5*2=10)	CO Mapping
a)	Explain how to convert the NFA to DFA and find the equivalent DFA for the following NFA.	5	CO 2
b)	Minimize the following DFA and explain the procedure followed.	5	CO 2
c)	Define Pumping Lemma for Regular Languages. Find out whether the following language is Regular or Not. $L = \{a^n b^n / n \geq 1\}$ over $\Sigma = \{a,b\}$.	5	CO 2
	UNIT II		
Q3	**Attempt any two parts**	(5*2=10)	CO Mapping
a)	Design a Pushdown Automata (PDA) for the following Language: $L = \{0^n 1^n / n \geq 1\}$ over $\Sigma = \{0,1\}$.	5	CO 2
b)	Convert the following CFG to CNF. $S \rightarrow ABA,\ A \rightarrow aA \mid \varepsilon,\ B \rightarrow bB \mid \varepsilon$	5	CO 2
c)	What is a derivation tree, discuss different types of derivation trees. Using the following grammar productions, find out the leftmost derivation and rightmost derivation for the string id+id*id. $A \rightarrow A+A$ $A \rightarrow A*A$ $A \rightarrow id$	5	CO 1

	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	q_0
q_1	q_2	$\{q_1, q_3\}$
$*q_2$	q_2	q_1
q_3	q_3	q_3

	0	1
$\rightarrow q_0$	q_1	q_5
q_1	q_6	q_2
$*q_2$	q_0	q_2
q_3	q_2	q_6
q_4	q_7	q_5
q_5	q_2	q_6
q_6	q_6	q_4
q_7	q_6	q_2